

```

}
}
libraryDB_RollNo> db.books_RollNo.insertMany([
  {book_id:101, title:"DBMS Concepts", author:"Korth", category:"Academics", price:550, copies_availabcopies_available:4},
  {book_id:102, title:"Operating Systems", author:"Silberschatz", category:"Academics", price:600, copcopies_available:3},
  {book_id:103, title:"Atomic Habits", author:"James Clear", category:"Self-Help", price:450, copies_acopies_available:5},
  {book_id:104, title:"Clean Code", author:"Robert C. Martin", category:"Programming", price:700, copicopies_available:2},
  {book_id:105, title:"Data Science 101", author:"Joel Grus", category:"Programming", price:650, copiecopies_available:3},
  {book_id:106, title:"Psychology Today", author:"Various", category:"Magazine", price:300, copies_avacopies_available:6}
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699bf6550a909470767c290d'),
    '1': ObjectId('699bf6550a909470767c290e'),
    '2': ObjectId('699bf6550a909470767c290f'),
    '3': ObjectId('699bf6550a909470767c2910'),
    '4': ObjectId('699bf6550a909470767c2911'),
    '5': ObjectId('699bf6550a909470767c2912')
  }
}
libraryDB_RollNo> db.members_RollNo.find()
[
  {
    _id: ObjectId('699bf6480a909470767c2907'),
    member_id: 1,
    name: 'Amit',
    age: 21,
    membership_type: 'Student',
    city: 'Kolkata',
    books_issued: 2,
    fine_amount: 0
  },
  {
    _id: ObjectId('699bf6480a909470767c2908'),
    member_id: 2,
    name: 'Neha',

```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>mongosh
Current Mongosh Log ID: 699bf4940a909470767c2906
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:      8.2.5
Using Mongosh:       2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-02-23T10:36:30.905+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test>

test> use libraryDB_RollNo
switched to db libraryDB_RollNo
libraryDB_RollNo> db.createCollection("members_RollNo")
| db.createCollection("books_RollNo")
{ ok: 1 }
libraryDB_RollNo> db.members_RollNo.insertMany([
| {member_id:1, name:"Amit", age:21, membership_type:"Student", city:"Kolkata", books_issued:2, fine_amfine_amount:0},
| {member_id:2, name:"Neha", age:24, membership_type:"Faculty", city:"Delhi", books_issued:3, fine_amofine_amount:50},
| {member_id:3, name:"Ravi", age:20, membership_type:"Student", city:"Mumbai", books_issued:1, fine_amfine_amount:0},
| {member_id:4, name:"Pooja", age:35, membership_type:"Faculty", city:"Chennai", books_issued:4, fine_fine_amount:100},
| {member_id:5, name:"Karan", age:22, membership_type:"Student", city:"Kolkata", books_issued:2, fine_fine_amount:20},
| {member_id:6, name:"Sneha", age:28, membership_type:"Research", city:"Delhi", books_issued:5, fine_afine_amount:0}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699bf6480a909470767c2907'),
    '1': ObjectId('699bf6480a909470767c2908'),
    '2': ObjectId('699bf6480a909470767c2909'),
  }
}
```

```
} ,
{
  _id: ObjectId('699bf6480a909470767c2909'),
  member_id: 3,
  name: 'Ravi',
  age: 20,
  membership_type: 'Student',
  city: 'Mumbai',
  books_issued: 1,
  fine_amount: 0
},
```

```
{
  _id: ObjectId('699bf6480a909470767c290b'),
  member_id: 5,
  name: 'Karan',
  age: 22,
  membership_type: 'Student',
  city: 'Kolkata',
  books_issued: 2,
  fine_amount: 20
}
```

```
}
```

```
]
```

```
libraryDB_RollNo> db.members_RollNo.find({books_issued:{$gt:2}})
```

```
[
```

```
{
```

```
  _id: ObjectId('699bf6480a909470767c2908'),
  member_id: 2,
  name: 'Neha',
  age: 24,
  membership_type: 'Faculty',
  city: 'Delhi',
  books_issued: 3,
  fine_amount: 50
},
```

```
{
```

```
  _id: ObjectId('699bf6480a909470767c290a'),
  member_id: 4,
```

```
books_issued: 2,
fine_amount: 20
},
{
  _id: ObjectId('699bf6480a909470767c290c'),
  member_id: 6,
  name: 'Sneha',
  age: 28,
  membership_type: 'Research',
  city: 'Delhi',
  books_issued: 5,
  fine_amount: 0
}
]
libraryDB_RollNo> db.members_RollNo.find({}, {name:1, membership_type:1, _id:0})
[
  { name: 'Amit', membership_type: 'Student' },
  { name: 'Neha', membership_type: 'Faculty' },
  { name: 'Ravi', membership_type: 'Student' },
  { name: 'Pooja', membership_type: 'Faculty' },
  { name: 'Karan', membership_type: 'Student' },
  { name: 'Sneha', membership_type: 'Research' }
]
libraryDB_RollNo> db.members_RollNo.find({membership_type:"Student"})
[
  {
    _id: ObjectId('699bf6480a909470767c2907'),
    member_id: 1,
    name: 'Amit',
    age: 21,
    membership_type: 'Student',
    city: 'Kolkata',
    books_issued: 2,
    fine_amount: 0
  },
  {
    _id: ObjectId('699bf6480a909470767c2909'),
```

```
_id: ObjectId('699bf6480a909470767c2908'),
member_id: 2,
name: 'Neha',
age: 24,
membership_type: 'Faculty',
city: 'Delhi',
books_issued: 3,
fine_amount: 50
},
{
  _id: ObjectId('699bf6480a909470767c2909'),
  member_id: 3,
  name: 'Ravi',
  age: 20,
  membership_type: 'Student',
  city: 'Mumbai',
  books_issued: 1,
  fine_amount: 0
},
{
  _id: ObjectId('699bf6480a909470767c290a'),
  member_id: 4,
  name: 'Pooja',
  age: 35,
  membership_type: 'Faculty',
  city: 'Chennai',
  books_issued: 4,
  fine_amount: 100
},
{
  _id: ObjectId('699bf6480a909470767c290b'),
  member_id: 5,
  name: 'Karan',
  age: 22,
  membership_type: 'Student',
  city: 'Kolkata',
  books_issued: 2,
```

```
{
  "_id": ObjectId("699bf6480a909470767c290b"),
  "member_id": 5,
  "name": "Karan",
  "age": 22,
  "membership_type": "Student",
  "city": "Kolkata",
  "books_issued": 2,
  "fine_amount": 20
},
{
  "_id": ObjectId("699bf6480a909470767c2907"),
  "member_id": 1,
  "name": "Amit",
  "age": 21,
  "membership_type": "Student",
  "city": "Kolkata",
  "books_issued": 2,
  "fine_amount": 0
},
{
  "_id": ObjectId("699bf6480a909470767c2909"),
  "member_id": 3,
  "name": "Ravi",
  "age": 20,
  "membership_type": "Student",
  "city": "Mumbai",
  "books_issued": 1,
  "fine_amount": 0
},
{
  "_id": ObjectId("699bf6480a909470767c290c"),
  "member_id": 6,
  "name": "Sneha",
  "age": 28,
  "membership_type": "Research",

```

```

    },
    {
      _id: ObjectId('699bf6480a909470767c290c'),
      member_id: 6,
      name: 'Sneha',
      age: 28,
      membership_type: 'Research',
      city: 'Delhi',
      books_issued: 5,
      fine_amount: 0
    }
  ]
libraryDB_RollNo> db.members_RollNo.find().sort({fine_amount:-1})
[
  {
    _id: ObjectId('699bf6480a909470767c290a'),
    member_id: 4,
    name: 'Pooja',
    age: 35,
    membership_type: 'Faculty',
    city: 'Chennai',
    books_issued: 4,
    fine_amount: 100
  },
  {
    _id: ObjectId('699bf6480a909470767c2908'),
    member_id: 2,
    name: 'Neha',
    age: 24,
    membership_type: 'Faculty',
    city: 'Delhi',
    books_issued: 3,
    fine_amount: 50
  },
  {
    _id: ObjectId('699bf6480a909470767c290b'),
    member_id: 5,

```

```
{
  _id: ObjectId('699bf6480a909470767c290a'),
  member_id: 4,
  name: 'Pooja',
  age: 35,
  membership_type: 'Faculty',
  city: 'Chennai',
  books_issued: 4,
  fine_amount: 100
},
{
  _id: ObjectId('699bf6480a909470767c290c'),
  member_id: 6,
  name: 'Sneha',
  age: 28,
  membership_type: 'Research',
  city: 'Delhi',
  books_issued: 5,
  fine_amount: 0
}
```

```
]
libraryDB_RollNo> db.members_RollNo.find({city:"Delhi"})
[
  {
    _id: ObjectId('699bf6480a909470767c2908'),
    member_id: 2,
    name: 'Neha',
    age: 24,
    membership_type: 'Faculty',
    city: 'Delhi',
    books_issued: 3,
    fine_amount: 50
  },
  {
    _id: ObjectId('699bf6480a909470767c290c'),
    member_id: 6,
    name: 'Sneha',

```

```
category: 'Programming',
price: 650,
copies_available: 3
}
```

```
]
libraryDB_RollNo> db.members_RollNo.aggregate([
| {$group:{_id:"$membership_type", avgFine:{$avg:"$fine_amount"}}}
| ])
[
```

```
{ _id: 'Research', avgFine: 0 },
{ _id: 'Student', avgFine: 6.666666666666667 },
{ _id: 'Faculty', avgFine: 75 }
]
```

```
libraryDB_RollNo> db.members_RollNo.updateMany(
| {membership_type:"Student"},
| {$set:{fine_amount:0}}
| )
{
```

```
acknowledged: true,
insertedId: null,
matchedCount: 3,
modifiedCount: 1,
upsertedCount: 0
}
```

```
libraryDB_RollNo> db.members_RollNo.aggregate([
| {$group:{_id:"$membership_type", avgFine:{$avg:"$fine_amount"}}}
| ])
[
```

```
{ _id: 'Faculty', avgFine: 75 },
{ _id: 'Student', avgFine: 0 },
{ _id: 'Research', avgFine: 0 }
]
```

```
libraryDB_RollNo> db.members_RollNo.updateMany(
| {membership_type:"Student"},
| {$set:{fine_amount:0}}
| )
{
```

```
acknowledged: true,
insertedId: null,
matchedCount: 3,
modifiedCount: 0,
upsertedCount: 0
}
```

```
libraryDB_RollNo> db.members_RollNo.aggregate([
| {$group:{_id:"$membership_type", avgFine:{$avg:"$fine_amount"}}},
| {$match:{avgFine:{$gt:30}}}
| ])
[ { _id: 'Faculty', avgFine: 75 } ]
```

```
libraryDB_RollNo> |
```

mongosh mongod://127.0.0.1 X + v

```
_id: ObjectId('699bf6480a909470767c290c'),
member_id: 6,
name: 'Sneha',
age: 28,
membership_type: 'Research',
city: 'Delhi',
books_issued: 5,
fine_amount: 0
}
]
libraryDB_RollNo> db.books_RollNo.find({price:{$gt:500}})
[
  {
    _id: ObjectId('699bf6550a909470767c290d'),
    book_id: 101,
    title: 'DBMS Concepts',
    author: 'Korth',
    category: 'Academics',
    price: 550,
    copies_available: 4
  },
  {
    _id: ObjectId('699bf6550a909470767c290e'),
    book_id: 102,
    title: 'Operating Systems',
    author: 'Silberschatz',
    category: 'Academics',
    price: 600,
    copies_available: 3
  },
  {
    _id: ObjectId('699bf6550a909470767c2910'),
    book_id: 104,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 700,
    copies_available: 2
  },
  {
    _id: ObjectId('699bf6550a909470767c2911'),
    book_id: 105,
    title: 'Data Science 101',
    author: 'Joel Grus',
    category: 'Programming',
    price: 650,
    copies_available: 3
  }
]
libraryDB_RollNo> db.members_RollNo.aggregate([
  {$group: {_id: "$membership_type", avgFine: {$avg: "$fine_amount"}}}]
```